

Apple

As a Product Analyst on the iPhone Camera team, you are tasked with evaluating camera quality metrics across different user groups....

Question 1: For the month of July 2024, how do average photo capture quality scores vary across different user groups? This analysis will help us identify segments where camera performance improvements could be prioritized.

```
SELECT f.user_group_id, user_group_name, AVG(photo_quality_score) AS avg_score
FROM fct_capture_quality AS f
INNER JOIN dim_user_group AS d
ON f.user_group_id = d.user_group_id
WHERE capture_date >= '2024-07-01' AND capture_date < '2024-08-01'
GROUP BY f.user_group_id, user_group_name
ORDER BY 3 DESC
```



Apple

As a Product Analyst on the iPhone Camera team, you are tasked with evaluating camera quality metrics across different user groups....

Question 2: Between August 1st and August 7th, 2024, how do photo capture quality scores change for each iPhone user group?
Use a 3-day average rolling window for this analysis.

```
SELECT f.user_group_id, user_group_name, capture_date,  
       AVG(photo_quality_score) OVER(PARTITION BY f.user_group_  
id ORDER BY capture_date ROWS BETWEEN 2 PRECEDING AND CURRENT  
ROW) AS rolling_avg_photo_score  
FROM fct_capture_quality AS f  
INNER JOIN dim_user_group AS d  
ON f.user_group_id = d.user_group_id  
WHERE capture_date >= '2024-08-01' AND capture_date <= '2024-0  
8-07'
```



Apple

As a Product Analyst on the iPhone Camera team, you are tasked with evaluating camera quality metrics across different user groups....

Question 3: For Q3 2024 (July 1st to September 30th), how do video capture quality scores trend for each iPhone user group when comparing each capture to the subsequent one? This insight will support our approach to optimize video performance through targeted hardware or software improvements.

```
SELECT f.user_group_id, user_group_name, video_quality_score,
       capture_date,
       LEAD(video_quality_score) OVER(PARTITION BY f.user_group
_id ORDER BY capture_date) AS next_video_quality_score,
       LEAD(video_quality_score) OVER(PARTITION BY f.user_group
_id ORDER BY capture_date) - video_quality_score AS score_tren
d
FROM fct_capture_quality AS f
INNER JOIN dim_user_group AS d
ON f.user_group_id = d.user_group_id
WHERE capture_date >= '2024-07-01' AND capture_date < '2024-10
-01'
ORDER BY f.user_group_id, capture_date
```

